

Docker Website Hosting Using Docker and Nginx

Step 1: Docker Desktop Installation and WSL Setup

Docker Desktop was installed successfully and WSL2 integration was enabled for Linux container support.



Step 2: Verifying Docker Installation

The docker --version command was executed in VS Code terminal to verify Docker installation.



Step 3: Checking Docker Engine

The docker ps command was executed to verify that Docker Engine was running correctly.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vigne>docker --version
Docker version 29.4.3, build 055a478

C:\Users\vigne>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
C:\Users\vigne>
```

Step 4: Running Hello World Container

The hello-world Docker container was executed successfully to test Docker functionality.

```
4. The Docker daemon streamed that output to the Docker client, which sent it
to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

C:\Users\vigne>
```

Step 5: Viewing Docker Images and Containers

The docker images and docker ps -a commands were used to view images and containers.

```
For more examples and ideas, visit:
https://docs.docker.com/get-started/

C:\Users\vigne>docker images
IMAGE           ID                DISK USAGE  CONTENT SIZE  EXTRA
hello-world:latest  0e760dfbc48      25.9kB      9.49kB        U

C:\Users\vigne>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
479ebd90a519   hello-world  "/hello"   About a minute ago  Exited (0) About a minute ago  frosty_kalam

C:\Users\vigne>
```

Step 6: Creating Ubuntu Linux Container

An Ubuntu container was created using interactive mode with the command docker run -it ubuntu bash.

```
C:\Users\vigne>docker run -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607dd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@c874e54bbc14:/#
```

Step 7: Running Linux Commands in Ubuntu

Basic Linux commands such as ls, pwd, and uname -a were executed inside the Ubuntu container.

```
C:\Users\vigne>docker run -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@c874e54bbc14:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@c874e54bbc14:/# pwd
/
root@c874e54bbc14:/# uname -a
Linux c874e54bbc14 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux
root@c874e54bbc14:/#
```

Step 8: Creating File Inside Ubuntu Container

A text file named myfile.txt was created inside the Ubuntu container using the touch command.

```
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@c874e54bbc14:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@c874e54bbc14:/# pwd
/
root@c874e54bbc14:/# uname -a
Linux c874e54bbc14 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux
root@c874e54bbc14:/# touch myfile.txt
root@c874e54bbc14:/# ls
bin boot dev etc home lib lib64 media mnt myfile.txt opt proc root run sbin srv sys tmp usr var
root@c874e54bbc14:/#
```

Step 9: Checking All Docker Containers

The docker ps -a command displayed all running and stopped containers.

```
C:\Users\vigne>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS          NAMES
c874e54bbc14   ubuntu        "bash"                  6 minutes ago   Exited (0) About a minute ago           intelligent_yonath
479ebd90a519   hello-world   "/hello"                9 minutes ago   Exited (0) 9 minutes ago           frosty_kalam

C:\Users\vigne>
```

Step 10: Docker Command Error Inside Container

An attempt to run Docker commands inside the Ubuntu container generated a command not found error.

```
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
root@B33a49694d5cd:/# docker ps
bash: docker: command not found
root@B33a49694d5cd:/#
```

Step 11: Exiting Ubuntu Container

The exit command was used to leave the Ubuntu container and return to Windows terminal.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
root@33a49694d5cd:/# docker ps
bash: docker: command not found
root@33a49694d5cd:/# exit
exit

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"

C:\Users\vigne>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
C:\Users\vigne>
```

Step 12: Docker Container Management

Docker containers were managed using docker ps and Docker Desktop interface.

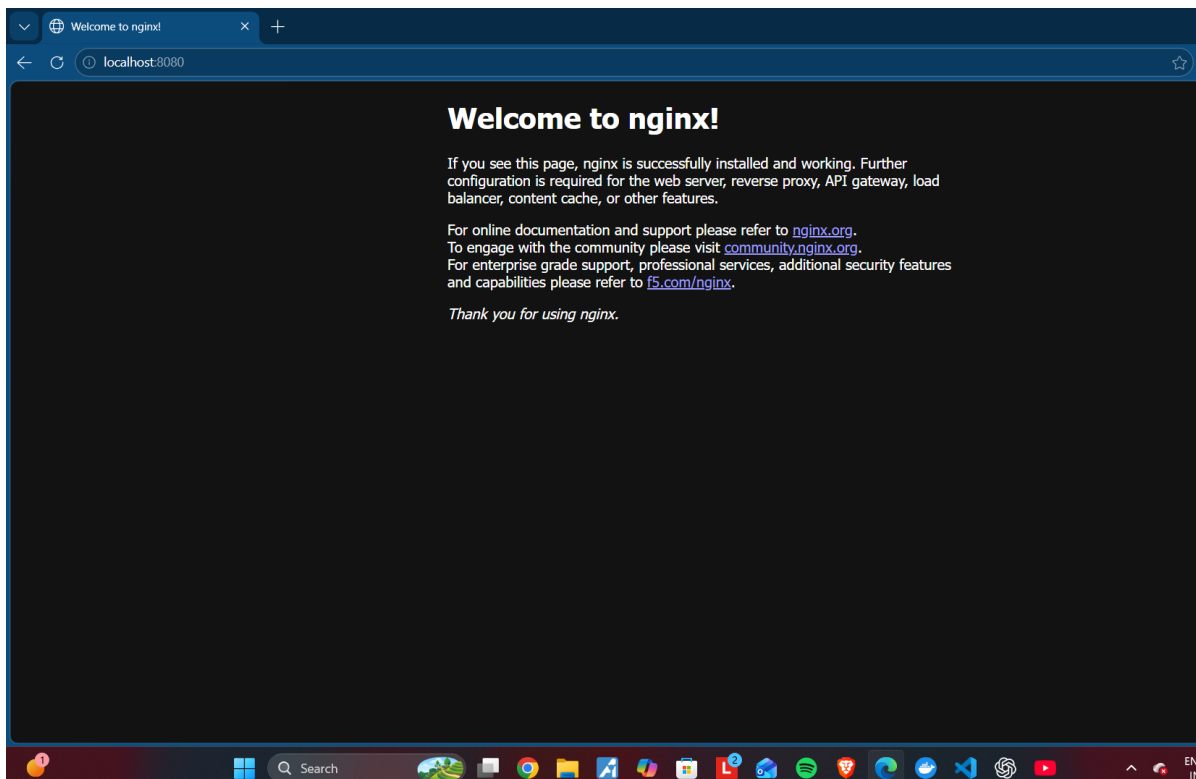
```
root@33a49694d5cd:/# docker ps
bash: docker: command not found
root@33a49694d5cd:/# exit
exit

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"

C:\Users\vigne>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
C:\Users\vigne>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
33a49694d5cd   ubuntu    "bash"    2 minutes ago   Exited (127) 53 seconds ago   myubuntu
c874e54bbc14   ubuntu    "bash"    11 minutes ago   Exited (0) 6 minutes ago     intelligent_yonath
479ebd90a519   hello-world  "/hello"  14 minutes ago   Exited (0) 14 minutes ago     frosty_kalam
C:\Users\vigne>
```

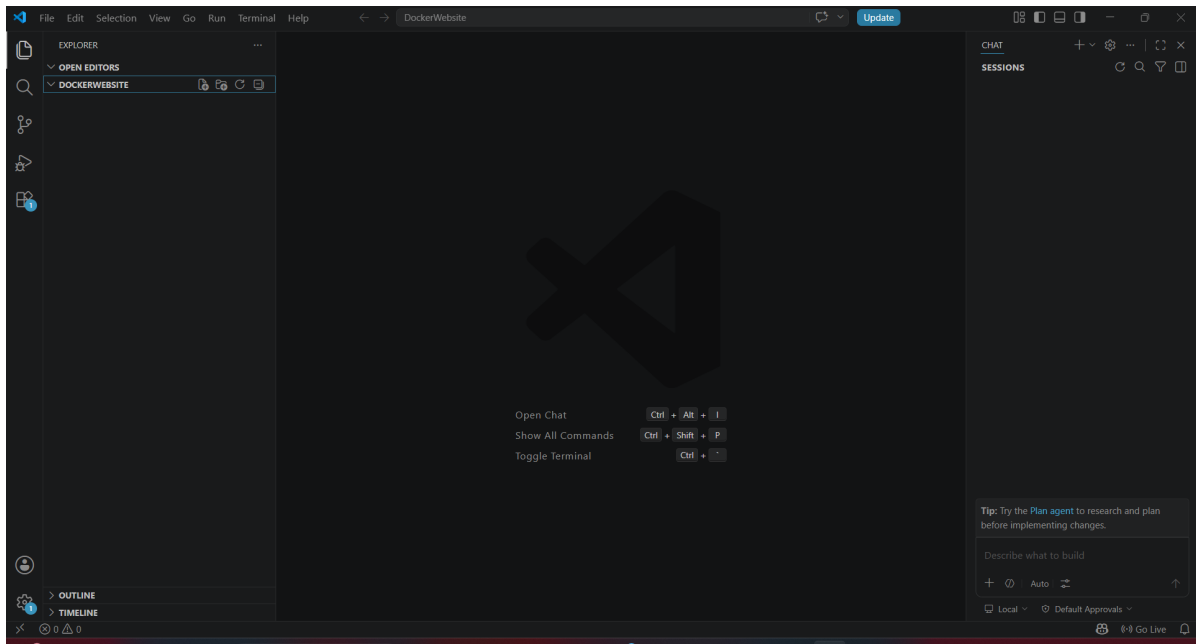
Step 13: Running Nginx Web Server

An Nginx container was launched successfully and accessed using localhost in the browser.



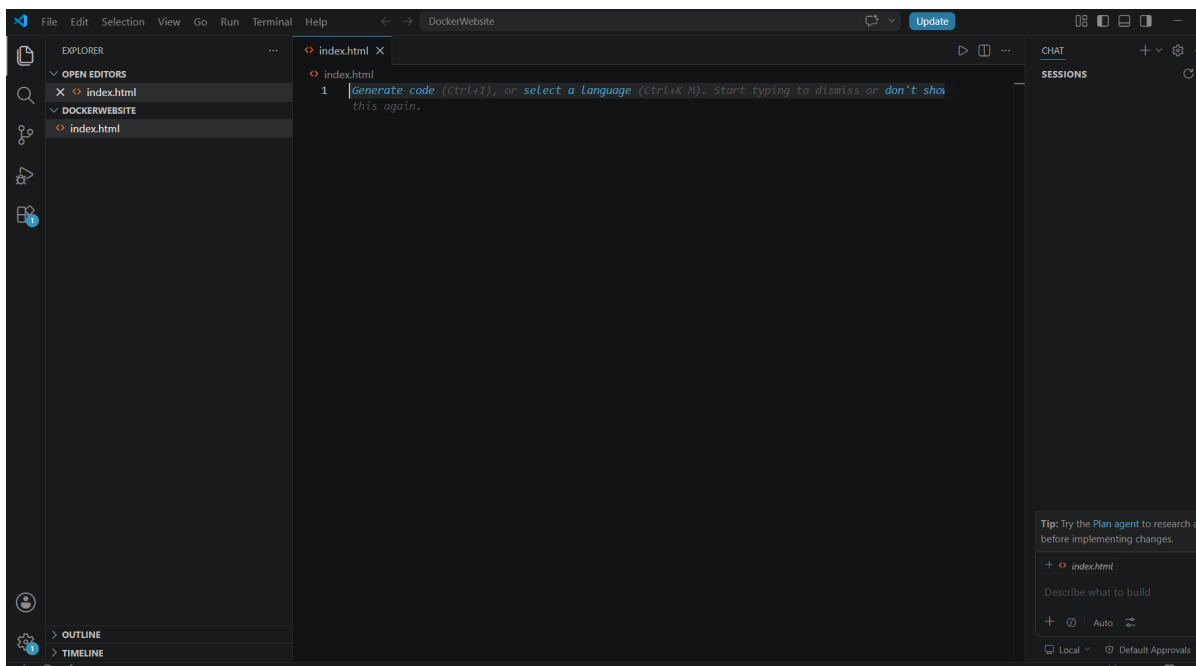
Step 14: Creating DockerWebsite Folder

A new project folder named DockerWebsite was created in Visual Studio Code.



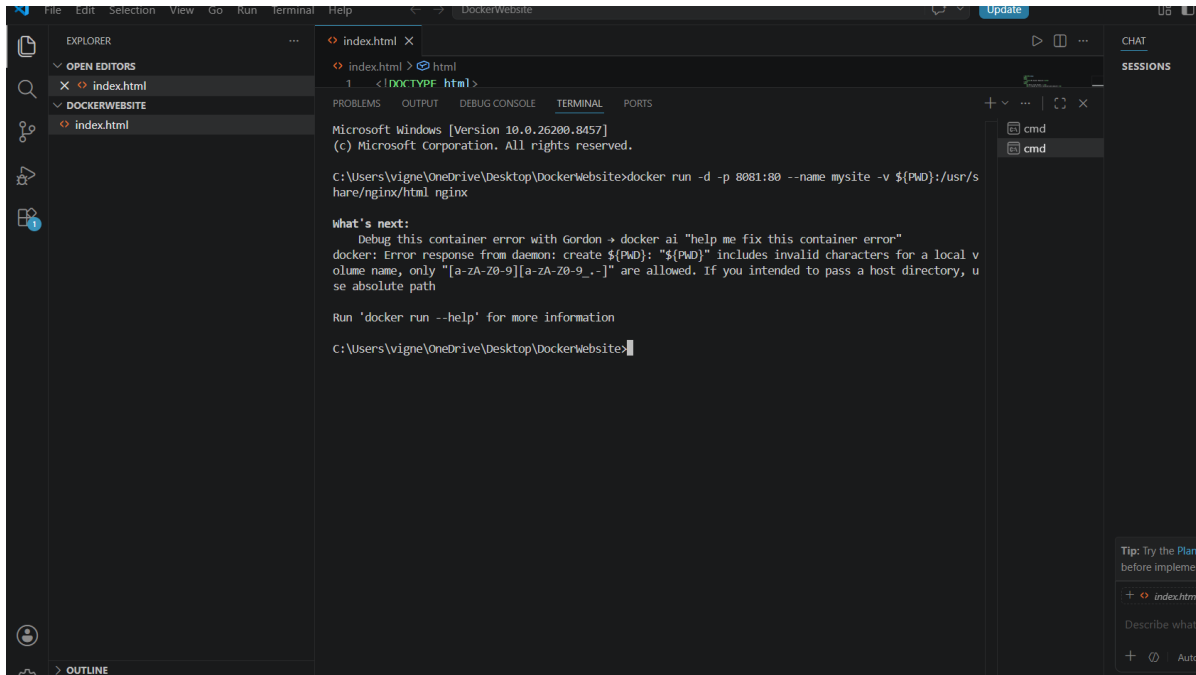
Step 15: Creating index.html File

An HTML file named index.html was created for the custom website.



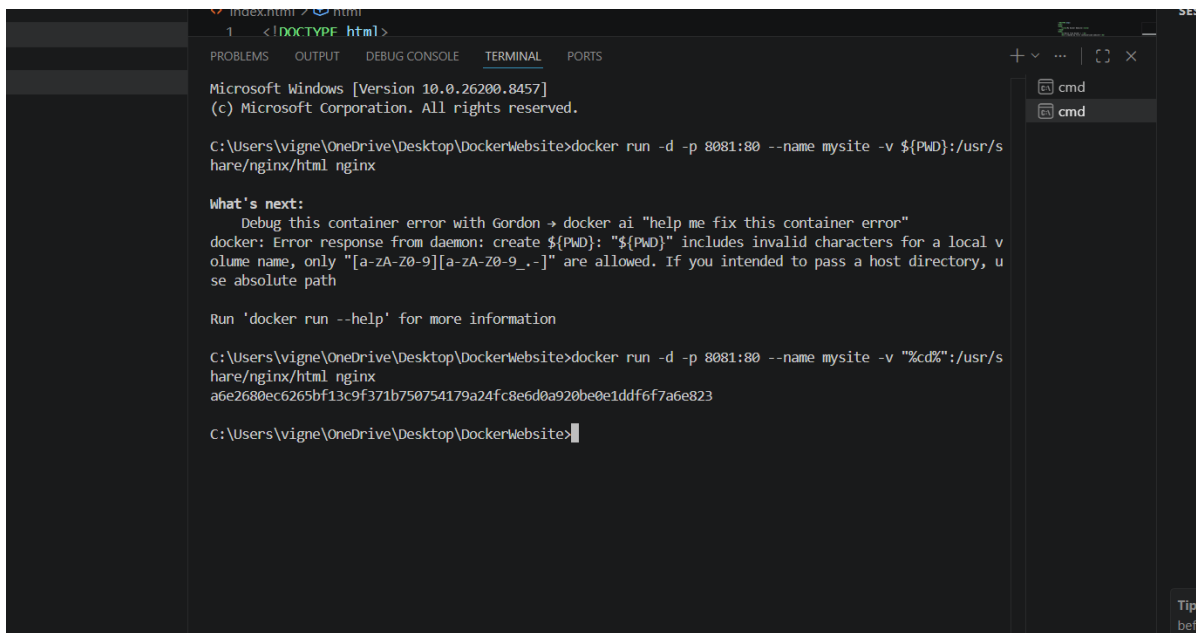
Step 16: Incorrect Docker Volume Mapping Command

The initial Docker run command failed because Linux-style \${PWD} syntax was unsupported in CMD.



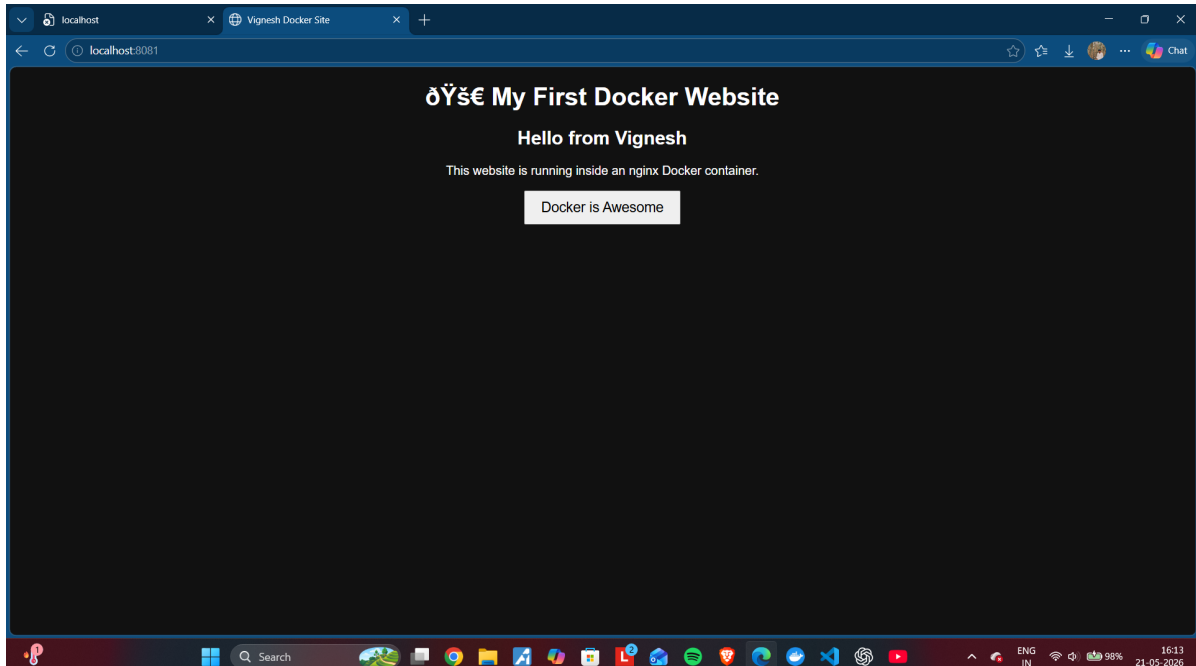
Step 17: Correct Docker Run Command

The Docker command was corrected using %cd% for proper volume mapping in Windows CMD.



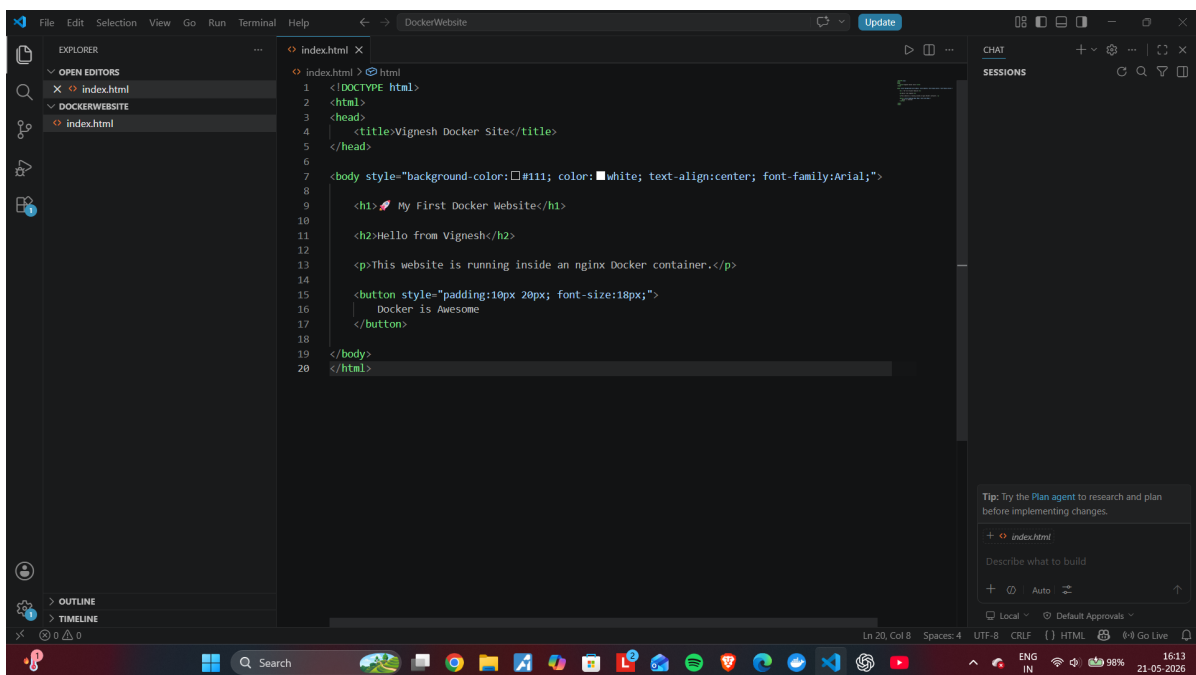
Step 18: Custom Website Hosted Successfully

The custom HTML website was successfully hosted using Docker and Nginx.



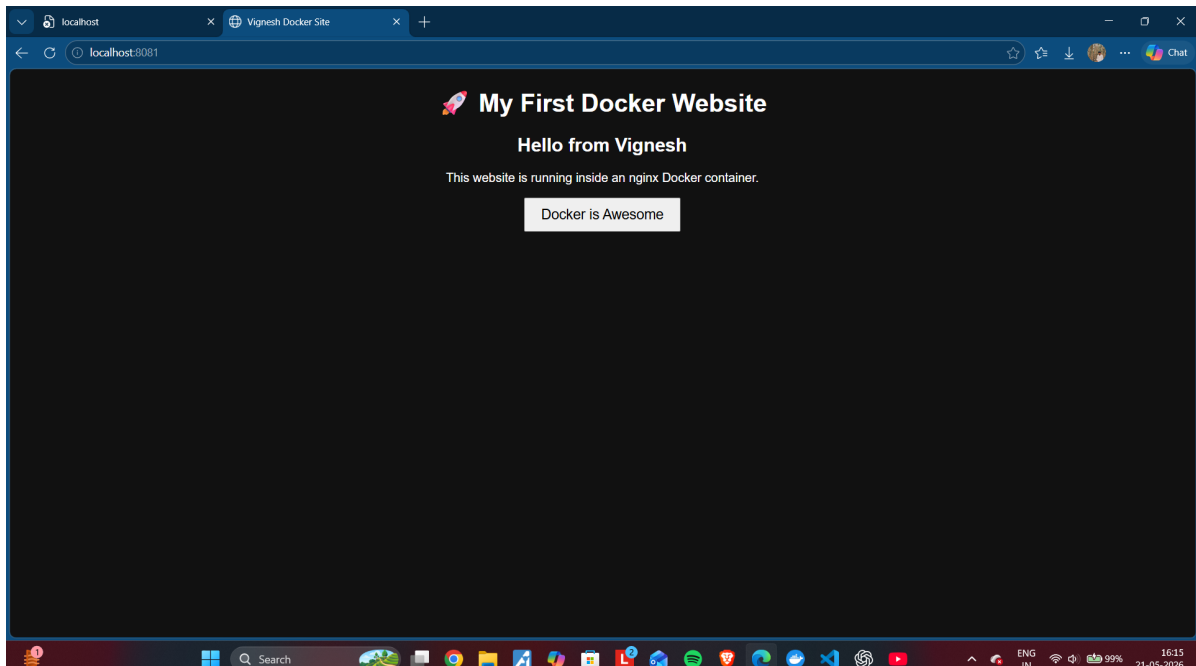
Step 19: HTML Source Code

The source code of the custom website was edited using Visual Studio Code.



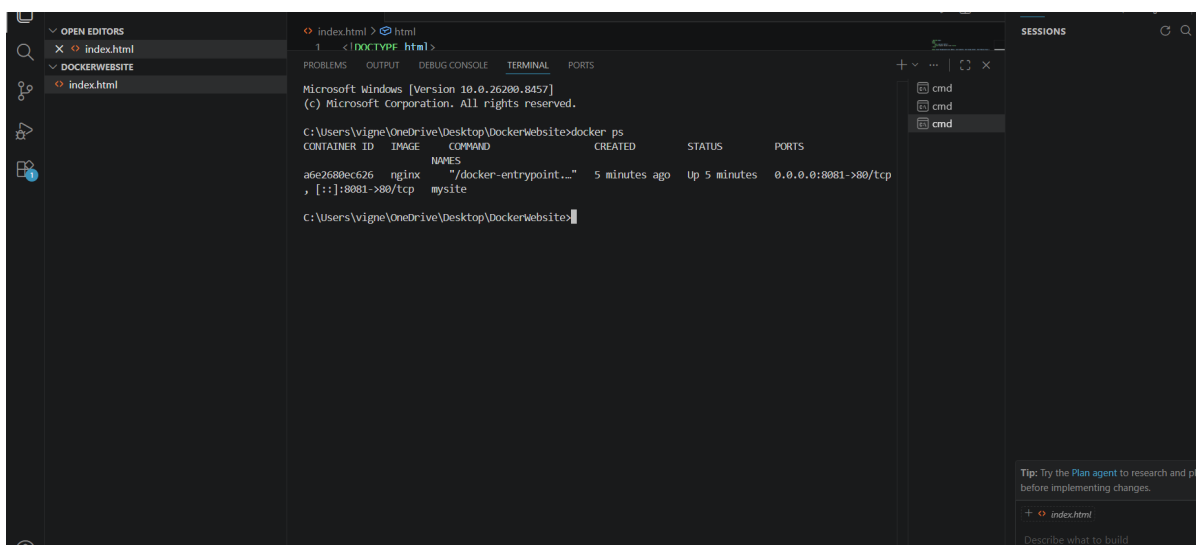
Step 20: Checking Running Website Container

The docker ps command confirmed that the mysite container was running successfully.



Step 21: Stopping Docker Containers

The docker stop command was used to stop active containers safely.



Docker Commands Used

```
docker --version
docker ps
docker run hello-world
docker images
docker ps -a
docker run -it ubuntu bash
ls
pwd
uname -a
touch myfile.txt
docker run -d -p 8080:80 --name mynginx nginx
docker run -d -p 8081:80 --name mysite -v "%cd%":"/usr/share/nginx/html nginx
docker stop mysite
docker stop mynginx
```

Conclusion: This project successfully demonstrated Docker installation, container creation, Linux container management, Nginx hosting, custom website deployment, port mapping, and volume mapping using Docker Desktop and Visual Studio Code.